Manan Sharma

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TECHNICAL SKILLS

Programming: Python, C, C++, Rust, R, TypeScript, JavaScript, Kotlin, MATLAB, SQL, Bash/Shell, PowerShell

AI / ML & Vision: PyTorch, TensorFlow, Keras, Intel OpenVINO, OpenCV, YOLOv8, DeepLabV3+, Residual U-Net (CRUNet), Scikit-learn, Model Context Protocol (MCP), Hugging Face, LLM Pipelines

Embedded & IoT: ESP32, OpenWrt, Linux Kernel Customization, Device Tree Overlays (.dts), SPI / I2C / UART Drivers, LoRaWAN (SX1303 Baseband), ChirpStack, MIPS (MT7628), ARM (NanoPi Allwinner H3), Raspberry Pi, Arduino

Systems & Cloud: Linux Administration (Debian, Ubuntu, Arch), Docker, Git, CI/CD, Azure, GCP, AWS (ECS, S3, Lambda), TCP/IP Networking, iptables, Network Security, DBMS

Full Stack & Scientific: FastAPI, Flask, React.js, Three.js (WebGL, GLSL Shaders), HTML5/CSS3, Tailwind CSS, NumPy, SciPy, MNE-Python (DSP / EEG), QGIS, Leaflet, ReportLab, Figma

EDUCATION

University of Jammu — Jammu & Kashmir, India
Bachelor of Technology (B.Tech) in Artificial Intelligence & Mathematics — *2024 – Present*

- **Relevant Coursework:** Artificial Intelligence, Machine Learning, Data Structures & Algorithms, Systems Programming, Linear Algebra, Discrete Mathematics, Database Management Systems (DBMS), Operating Systems, Computer Networks.

Kendriya Vidyalaya No. 1, Udhampur — Jammu & Kashmir, India
Higher Secondary Education (Senior Secondary) — *Graduated 2024*

EXPERIENCE

Embedded & Internet of Things Engineer (Intern) – UniConverge Technologies — Jun. 2026 – Jul. 2026
Noida, Uttar Pradesh, India — *On-site*

- Built custom **embedded Linux firmware** images utilizing OpenWrt build environments, establishing seamless cross-compilation interoperability across **MIPS (MT7628)** and **ARM (NanoPi Allwinner H3)** gateway architectures.
- Engineered hardware integration for **Semtech SX1303 LoRa baseband concentrators** by authoring custom device tree overlays (`.dts`), pin multiplexing configurations, and kernel-space SPI drivers.
- Optimized **kernel-space SPI drivers** and tuned the Linux TCP/IP networking stack for deterministic high-throughput, low-latency edge packet transmission across distributed sensor nodes.
- Configured and deployed **ChirpStack IoT server stacks** and LoRaWAN packet forwarders for enterprise industrial telemetry and edge gateway fleet monitoring.
- Collaborated with engineering teams to automate board bring-up scripts and flashing workflows, cutting multi-board regression testing cycles by 40%.

Research Intern (Computer Vision & Edge AI) – Indian Institute of Technology (IIT), Roorkee — Jun. 2025 – Jul. 2025
Roorkee, Uttarakhand, India — *On-site*

- Researched and developed an optimized real-time **Object Detection pipeline** based on custom-trained **YOLOv8** architectures for sustained low-latency edge video analytics.
- Accelerated deep learning inference on **Intel Core Ultra Neural Processing Units (NPUs)** leveraging the **Intel OpenVINO toolkit**, converting models to INT8/FP16 Intermediate Representations (IR).
- Benchmarked multi-device throughput and latency curves across CPU, integrated GPU, and dedicated NPU execution units, achieving continuous 60+ FPS inference with minimal power draw and thermal footprint.
- Built internal data annotation, dataset augmentation, and automated validation pipelines for custom computer vision models with OpenCV.

Technical Organizer & Systems Lead – Innovathon & University Research — Aug. 2024 – Present
University of Jammu, India — *Leadership / Research*

- Core technical organizer for **Innovathon 1.0 National Hackathon**, leading platform operations, server infrastructure, and mentoring 200+ collegiate participants on edge computing and machine learning.
- Architected telemetry ingestion pipelines and open-source scientific computing templates adopted across departmental research groups.
- Selected for dramatic portrayal as Arjuna and Guru Kripacharya in *Rashmirathi* at the **25th Bharat Rang Mahotsav (BRM)** International Theatre Festival (2026) organized by the National School of Drama (NSD) & Ministry of Culture.

PROJECTS

OJASEWS — 3D Landslide Twin & Early-Warning | *Python, FastAPI, React 19, Three.js, ESP32, Random Forest, Docker* 2026 – Present

- Architected an autonomous geotechnical early-warning platform monitoring landslide precursors across steep Himalayan corridors (calibrated on NH-44 highway), handling synthetic 7-scenario physics simulation and live ESP32 IoT node telemetry.
- Engineered an interactive **Three.js 3D Himalayan digital twin** with procedural elevation/slope shaders, risk heatmap blending, live rainfall particles, and debris-flow simulation alongside a liquid-slosh risk gauge and synchronized Recharts analytics.
- Designed a **dual-track risk arbiter** coupling geotechnical physics heuristics (moisture trend, tilt velocity, infiltration lag) with a 4-class **Random Forest ML classifier** and automated **ntfy.sh** sub-second mobile push dispatch.

GeoChange AI — Satellite Environmental Intelligence | *Python, FastAPI, Sentinel-2 (10m), Landsat, NASA Carbon, Leaflet* 2026

- **Won Best Presentation Award at INDU X 5.0 International Hackathon** for building an end-to-end multi-spectral remote sensing platform detecting land-cover transformations across 5,500+ hectares (2021–2026).
- Enforced a strict **Data Readiness Gate** querying STAC providers (AWS Earth Search, Planetary Computer, CDSE) with sub-pixel alignment and automated Cloud Score+ / QA60 cloud-mask verification.
- Computed pixel-level **Year of Change maps**, quantified lost carbon storage in tonnes of CO2e via **NASA/ORNL Global Biomass layers**, measured Landsat 8/9 thermal heat island deltas, and generated ReportLab PDF reports.

NeuroView — 3D Spatial Telemetry & WebGL | *Python, FastAPI, Three.js, WebGL/GLSL, MNE, Groq, Docker* 2026 – Present

- Built an interactive **Three.js WebGL engine** with custom **GLSL shaders** executing real-time **Inverse Distance Weighting (IDW) scalar field heatmap interpolation** across arbitrary 3D surfaces.
- Engineered a high-throughput **FastAPI backend** utilizing **NumPy and MNE-Python** for parsing binary data streams, baseline noise filtering, and spatial electrode coordinate projection.
- Integrated an automated diagnostic pipeline utilizing **cascading LLMs (Groq API, DeepSeek)** for real-time anomaly detection and natural language telemetry reporting within a containerized Docker architecture.

SlipSense — Res U-Net Landslide Mapping | *PyTorch, DeepLabV3+, CRUNet, HDF5, QGIS, Flask* 2025 – Present

- Developed an AI-powered semantic segmentation system to delineate landslide hazards across Himalayan terrain using **14-band multispectral satellite imagery (Sentinel-2 HDF5)**.
- Designed a custom **Residual U-Net (CRUNet)** integrated with **Atrous Spatial Pyramid Pooling (ASPP)** from DeepLabV3+ to capture multi-scale spatial receptive fields and eliminate vanishing gradients.
- Exposed trained models via a high-throughput **Flask REST API**, enabling sub-second inference and automated hazard boundary mapping for disaster mitigation.

YOLO Live Object Detection on NPU | *Python, YOLOv8, Intel OpenVINO, NPU, OpenCV* 2025

- Quantized YOLOv8 weights to INT8/FP16 Intermediate Representation (IR) via **Intel OpenVINO**, deploying directly to **Intel Core Ultra NPUs** for sustained 60+ FPS real-time inference with minimal power and thermal footprint.

Custom Raspberry Pi Router & Network Hardening | *OpenWrt, Linux Kernel, iptables, Bash, DoH* 2024

- Compiled custom OpenWrt Linux firmware with hardened iptables routing policies, stateful packet inspection, NAT masquerading, and DNS-over-HTTPS (DoH).

CERTIFICATIONS & HONORS

Vikrant Univ. / ARINA AI / GO KLOUD — Best Presentation Award (INDU X 5.0 Int'l Hackathon) Issued: Nov. 2026

- **Skills:** Satellite Remote Sensing, Data Readiness Verification, Environmental AI, STAC Protocols, Technical Pitching

AMD — AI Agents 101: Building AI Agents with MCP & Open-Source Inference Issued: Aug. 2026

- **Skills:** Model Context Protocol (MCP), Agentic Workflows, Open-Source Inference, Tool Calling, System Architecture

UniConverge Technologies — Embedded & IoT Engineering Industrial Certification Issued: Jul. 2026

- **Skills:** Embedded Linux, OpenWrt, LoRaWAN, ChirpStack, Kernel Drivers, SPI Drivers, Hardware Prototyping

IIT Kanpur — Junior Scientist Honor (26th National Children Science Congress) Issued: Dec. 2018

- **Skills:** Scientific Research Methodology, Experimental Design, State Representative, Technical Presentation

Ministry of Education & AICTE — IDE Innovation & Entrepreneurship Bootcamp Issued: Jul. 2026

- **Skills:** Deep-Tech Venture Design, Rapid Prototyping, Product Architecture, Technology Commercialization